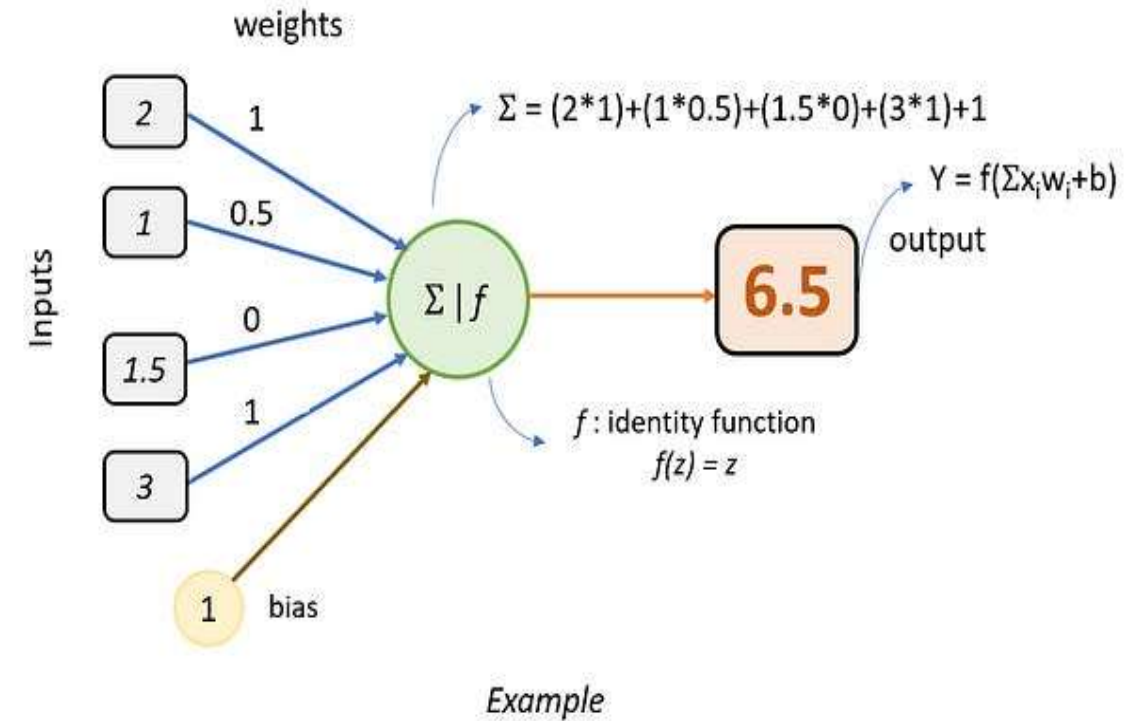
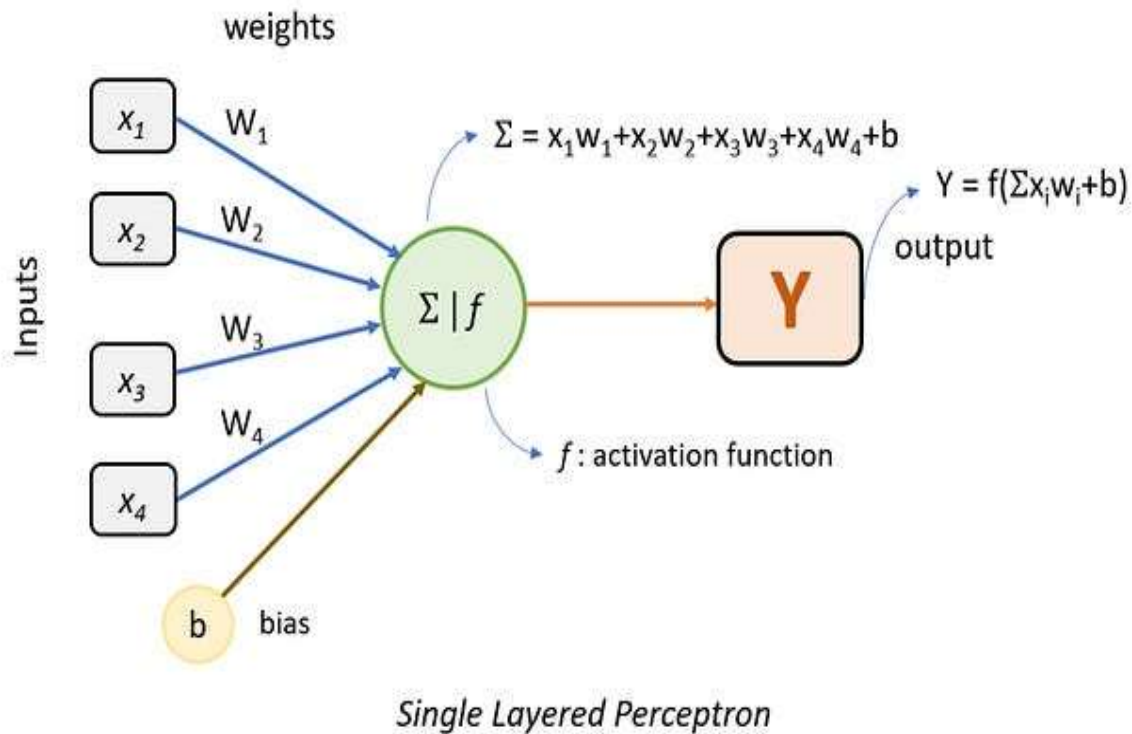


Backpropagation in Neural Network

Part 01

H.M. Samadhi Chathuranga Rathnayake

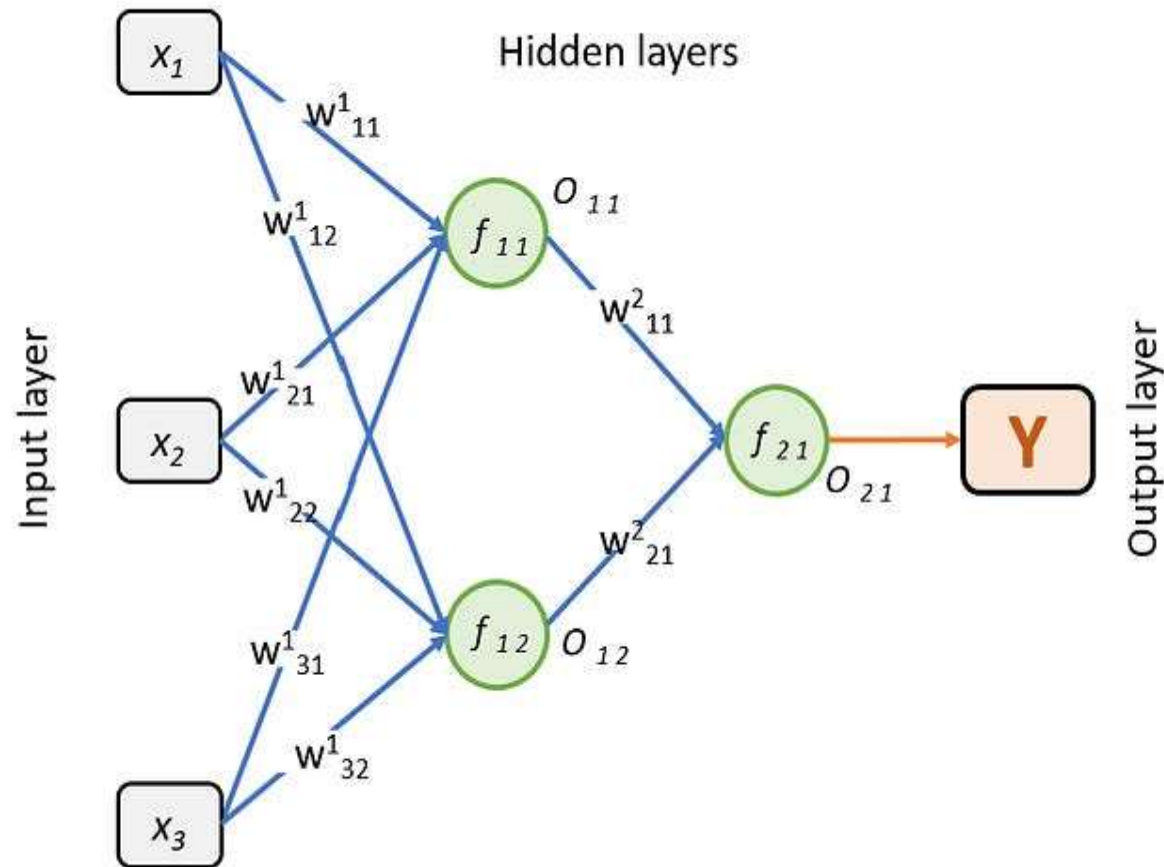
Single Layer Perceptron



Multi Layer Perceptron

- A Multi-Layered Perceptron NN can have n-number of hidden layers between input and output layer.
- These hidden layer can have n-number of neurons, in which the first hidden layer takes input from input layer and process them using activation function and pass them to next hidden layers until output layer.
- Every neuron in a hidden layer uses a non-linear activation function.

Multi Layer Perceptron



here,

x - inputs

w^n_{ij} - n is num of next hidden layer,

i is neuron num of previous layer,

j is neuron num of next layer

(w^2_{21} - n is next layer,

i is 2nd neuron of previous layer,

j is 1st neuron of next layer)

f_{ij} - i is num of layer and j is num of neuron

O_{ij} - output from i^{th} layer j^{th} neuron

Y - output

Data Preprocessing

- We cannot directly put any form of data into a Neural Network. The data we feed into a NN should be numerical(real values).
- If you have categorical data, it should be converted into numerical values using techniques like one-hot encoding.
- If a column has real values but ranges with huge difference(eg:1 to 10000), it should be scaled down using techniques like normalization, standardization.

Training MLP Networks

- A deep MLP neural network tries to learn underlying pattern or to map inputs and outputs using weights by training with given data.
- So, to achieve this we use a optimization technique called Gradient Descent.
- The weights and bias will be updated using a method called backpropagation.
- We calculate loss using a loss function and calculate the derivate and we update the weights during backpropagation.
- The main goal is to minimize the difference(loss) between predicted output and actual output.

Training MLP Example

X1	X2	X3	Y
1	4	7	10
2	5	8	15
3	6	9	20

Training MLP Example

- Consider a single layer MLP network.
- Consider the following conditions.

Linear activation: $f(z) = z$

MSE Loss: $L = \frac{1}{n} \sum (y_{true} - y_{pred})^2$

SGD: Update after each row

Training MLP Example

- Let:
 - Weights: w_1, w_2, w_3
 - Bias: b
 - Assume initial values:
 - $w_1=0.5, w_2=-0.5, w_3=1.0, b=0$
 - Learning rate: $\alpha=0.01$

Training MLP Example

- Row 1
- Forward Pass

$$z = w_1 \times x_1 + w_2 \times x_2 + w_3 \times x_3 + b = (0.5)(1) + (-0.5)(4) + (1.0)(7) + 0 = 0.5 - 2 + 7 = 5.5$$

- Prediction

$$\hat{y} = f(z) = z = 5.5$$

- Loss (MSE)

$$L = (y - \hat{y})^2 = (10 - 5.5)^2 = 20.25$$

Training MLP Example

- Backward Pass

$$\frac{\partial L}{\partial \hat{y}} = -2(y - \hat{y}) = -2(10 - 5.5) = -9.0$$

$$\hat{y} = w_1 \times x_1 + w_2 \times x_2 + w_3 \times x_3 + b$$

$$\frac{\partial \hat{y}}{\partial w_1} = x_1$$

- With the chain rule,

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial w_1} = (-2)(y - \hat{y}) \times x_1$$

Training MLP Example

- Others too

$$\frac{\partial L}{\partial w_2} = \frac{\partial L}{\partial \hat{y}} \times x_2$$

$$\frac{\partial L}{\partial w_3} = \frac{\partial L}{\partial \hat{y}} \times x_3$$

$$\frac{\partial L}{\partial b} = \frac{\partial L}{\partial \hat{y}} \times 1 = \frac{\partial L}{\partial \hat{y}}$$

Training MLP Example

- So for w_1 ,

$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial \hat{y}} \times x_1 = (-9.0)(1) = -9.0$$

$$\frac{\partial L}{\partial w_2} = (-9.0)(4) = -36.0$$

$$\frac{\partial L}{\partial w_3} = (-9.0)(7) = -63.0$$

$$\frac{\partial L}{\partial b} = -9.0$$

- With SGD

$$w_1 = w_1 - \alpha \times \frac{\partial L}{\partial w_1} = 0.5 - 0.01(-9.0) = 0.59$$

$$w_2 = -0.5 - 0.01(-36.0) = -0.14$$

$$w_3 = 1.0 - 0.01(-63.0) = 1.63$$

$$b = 0.0 - 0.01(-9.0) = 0.09$$

Training MLP Example

- Row 2

$$z = (0.59)(2) + (-0.14)(5) + (1.63)(8) + 0.09 = 1.18 - 0.7 + 13.04 + 0.09 = 13.61$$

$$\hat{y} = z = 13.61$$

$$L = (15 - 13.61)^2 = 1.9321$$

$$\frac{\partial L}{\partial \hat{y}} = -2(15 - 13.61) = -2(1.39) = -2.78$$

$$\frac{\partial L}{\partial w_1} = (-2.78)(2) = -5.56$$

$$\frac{\partial L}{\partial w_2} = (-2.78)(5) = -13.9$$

$$\frac{\partial L}{\partial w_3} = (-2.78)(8) = -22.24$$

$$\frac{\partial L}{\partial b} = -2.78$$

Training MLP Example

- Row 2

$$w_1 = 0.59 - 0.01(-5.56) = 0.6456$$

$$w_2 = -0.14 - 0.01(-13.9) = 0.0$$

$$w_3 = 1.63 - 0.01(-22.24) = 1.8524$$

$$b = 0.09 - 0.01(-2.78) = 0.1178$$

Training MLP Example

- Row 3

$$z = (0.6456)(3) + (0.0)(6) + (1.8524)(9) + 0.1178 = 1.9368 + 0 + 16.6716 + 0.1178 = 18.7262$$

$$L = (20 - 18.7262)^2 = 1.6185$$

$$\frac{\partial L}{\partial \hat{y}} = -2(20 - 18.7262) = -2(1.2738) = -2.5476$$

$$\frac{\partial L}{\partial w_1} = (-2.5476)(3) = -7.6428$$

$$\frac{\partial L}{\partial w_2} = (-2.5476)(6) = -15.2856$$

$$\frac{\partial L}{\partial w_3} = (-2.5476)(9) = -22.9284$$

$$\frac{\partial L}{\partial b} = -2.5476$$

Training MLP Example

- Row 3

$$w_1 = 0.6456 - 0.01(-7.6428) = 0.7220$$

$$w_2 = 0.0 - 0.01(-15.2856) = 0.1529$$

$$w_3 = 1.8524 - 0.01(-22.9284) = 2.0817$$

$$b = 0.1178 - 0.01(-2.5476) = 0.1433$$

Training MLP Example

- After the 1st epoch

$$w_1 = 0.7220$$

$$w_2 = 0.1529$$

$$w_3 = 2.0817$$

$$b = 0.1433$$